

STATEMENT OF PURPOSE

Applicant Name: Ayush Raj

CGPA: 8.93

Target Program: Amazon ML Summer School 2026

Date: June 14, 2026

My interest in Artificial Intelligence and Machine Learning comes from a desire to build applications that solve practical, everyday problems. To gain hands-on experience, I built **Smart Expense Tracker**, a full-stack Progressive Web App (PWA) using React, FastAPI, and PostgreSQL. I wanted to see how machine learning could make managing personal finance easier and more automated.

During this project, I worked with several key ML areas. First, I integrated Large Language Models (LLMs) via **Gemini and Groq APIs** to create a conversational assistant. I wrote custom system prompts to parse unstructured user chat messages into structured JSON schemas that match our PostgreSQL database models. Second, I set up a **Computer Vision (OCR)** pipeline to automatically extract transaction details from uploaded receipt photos. Third, I implemented a statistical **Z-Score anomaly detection algorithm** to flag unusual transactions. If a user's new transaction was more than 2.0 standard deviations away from their average spending, the app triggered a budget alert. While building this was a great learning experience, it made me realize the limits of using pre-built APIs. I had very little control over model accuracy, latency, and data privacy.

This project helped me identify the gaps in my current machine learning knowledge. I want to move beyond calling black-box APIs and learn how to train and fine-tune custom models myself. In computer vision, instead of using generic OCR endpoints, I want to learn to build and optimize custom Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to handle noisy, low-quality receipt images. On the data side, simple statistical Z-scores assume data is normally distributed, which isn't always true for real spending habits. I want to learn advanced unsupervised techniques like Isolation Forests or autoencoders to build better anomaly detectors. Finally, I need to learn how to manage large-scale datasets and deploy low-latency models on cloud platforms like AWS.

The Amazon ML Summer School (MLSS) 2026 is exactly what I need to bridge these gaps. Learning directly from Amazon's ML scientists about deep learning, vision, and NLP matches my goal of transitioning from a full-stack developer to a machine learning engineer. With a strong academic background (CGPA: 8.93) and practical development experience, I am ready to take this next step. I am confident this program will give me the mathematical foundation and engineering skills to build next-generation AI systems.